

Aegis Orbital Compute Node

1 MW Batch-Compute Node, Dawn-Dusk Sun-Synchronous Orbit

Architecture Reference Document | Rev B — September 2026 | Not for Hardware Commitment

1. Executive Summary

The Aegis Orbital Compute Node (AOCN) is a free-flying unit of one megawatt of commercial accelerator load in an unpressurised shielded vault, powered by a fixed 5,400 m² solar sheet and cooled by 2,000 m² of pumped-loop radiator, in a 650 km dawn-dusk sun-synchronous orbit. The orbit has no eclipse season, so the node carries no energy storage beyond a safe-mode pack and runs continuously. The node is sized for batch work: model training, large-scale inference, simulation, and processing of data already in orbit. It is not a cloud region.

Rev B rebuilds the sizing from the load up. Accelerator load is 1.0 MW IT with 1.2 MW electrical end of life. Solar area is stated end of life with derates shown. Water shielding is replaced by an aluminium vault and a 3–5 year tray refresh matched to the commercial hardware cycle. Node mass is ~65 t with 20 % growth. The document states the data path and the launch price at which the node reaches parity with a terrestrial hall: about \$300 per kilogram to sun-synchronous orbit. That is a launch-price gate. The node itself is buildable with current spaceflight technology.

2. Purpose and Positioning

The case, in three parts

- 1 **The terrestrial constraint is the grid.** Interconnect queues, cooling water, and siting permits gate AI capacity on Earth. None exist in orbit. Power is not free; it costs launched array mass. A square metre of array in a dawn-dusk orbit yields $\sim 4.5\times$ the annual energy of the same square metre on the ground, lit 100 % of the time at full solar constant with no storage.
- 2 **Batch work is comms-light.** Training is weeks of computation on a dataset that arrives once and a result that leaves once. The dataset is launched on solid-state storage; results return over optical links. Section 4 sizes the path.
- 3 **Some customers are already in space.** Earth-observation constellations and future cislunar operations want compute seconds away. This is the only case with no terrestrial competitor and today the smallest; AOCN is designed to serve it without being designed around it.

What AOCN is not. Not a cloud region, not an edge for latency-sensitive traffic, and not a claim that orbital compute is cheaper today. *At current launch prices it is two to five times the cost of a terrestrial hall for the same accelerators. Section 10 states where that crosses over.*

3. Orbit Selection

Parameter	Value
Orbit	650 km circular, sun-synchronous, ~98° inclination, dawn-dusk local time of ascending node
Period	~97.7 min
Sun angle to orbit normal	$\leq 23.4^\circ$ through the year
Eclipse	None at any season above ~620 km (Earth angular radius from 650 km is 65.2° , below the 66.6° minimum beta angle)
Energy storage	Safe-mode pack only, order 50 kWh
Array pointing	Fixed sheet in the orbit plane, facing the sun; no gimbal
Radiator pointing	Wings in the plane containing the sun vector and nadir; faces look along $\pm$ velocity, never at the sun, obliquely at Earth; shaded by the array
Drag makeup	~2,100 m ² ram area; ~13 m/s per year; < 50 kg xenon per year on electric propulsion at mean solar activity
Ground contact	Both poles every orbit; ~150 min per day of contact with a single high-latitude terminal
Penalty	Radiation: auroral horns and the South Atlantic Anomaly; total dose several times an ISS-orbit figure (Section 7)

Rev 1.1 assumed a generic 500–600 km orbit with a 35 % worst-case eclipse and 2.6 t of battery per MW. That figure was cell-level; at a LEO-survivable depth of discharge of ~35 % with pack overhead it is ~12 t per node. That is a sizing problem, not a barrier. Rev B chooses the orbit that does not need a battery because everything downstream is simpler: a fixed array, a constant sun, one radiator plane, no charge cycle every orbit.

4. Data Path

Leg	Method	Capacity	Note
Dataset up	Launched solid-state storage installed with each tray refresh	10 PB $\approx$ 164 $\times$ 61 TB drives, < 0.5 t	Cheaper and faster than any uplink
Results down	Optical to polar ground stations; Ka-band fallback	$\sim$ 110 TB/day at 100 Gb/s over $\sim$ 150 min of contact	1 TB checkpoint at 10 Gb/s: $\sim$ 13 min; station diversity covers weather
Tasking, telemetry	RF via relay, continuous	kb/s to Mb/s	Ground plans, node executes autonomously between contacts
Node to node	Optical crosslink within the plane	10–100 Gb/s	Distributed training across nodes; not required for one node

5. Thermal Architecture

At steady state every watt of electrical input becomes heat. Commercial accelerators hold junction temperature below $\sim$ 363 K; direct liquid cooling delivers coolant at 305–320 K and returns it 10–15 K warmer. The radiator runs between $\sim$ 330 K inlet and $\sim$ 305 K outlet: a 315 K mean radiating temperature. Rejection is two-sided at emissivity 0.88 with an effective factor of 0.6 covering Earth infrared on the oblique face, inter-wing view factor, coating degradation, and fin efficiency. With the wings shaded by the array and no direct solar load, the factor is conservative.

Mean radiating temperature	Two-sided, theoretical	Effective ($\times$ 0.6)	Area per MW rejected
300 K	808 W/m ²	485 W/m ²	2,060 m ²
310 K	922 W/m ²	553 W/m ²	1,810 m ²
315 K, baseline	983 W/m²	590 W/m²	1,700 m²
320 K	1,046 W/m ²	628 W/m ²	1,590 m ²
330 K	1,183 W/m ²	710 W/m ²	1,410 m ²

Parameter	Value
Heat rejected	1.2 MW
Radiator area	~2,000 m ² two-sided; two 50 × 20 m wings, zenith and nadir of the truss
Loop	Single-phase pumped, redundant pumps, isolable sections per wing
Panel mass	6 kg/m ² deployable pumped-loop panel; ~12 t
Coatings	Atomic-oxygen tolerant; ram-facing wing faces
Growth path	Hotter-tolerant accelerator generations raise rejection per m ² on the same wings

6. Power Architecture

Step	Factor	Running value
Solar constant 1,361 W/m ² × 29 % triple-junction GaAs	—	395 W/m ²
Blanket packing: cells, gaps, harness, hinges	0.85	336 W/m ²
Radiation degradation, 10 years polar	0.85	285 W/m ²
Operating temperature derate	0.85	243 W/m ²
Worst-season sun angle on a fixed array, cos 23.4°	0.92	223 W/m² end of life

Parameter	Value
Electrical demand, end of life	1.2 MW (1.0 MW IT plus ~20 % pumps, conversion, comms, attitude control, propulsion)
Array area	5,400 m ² , carried as a 90 × 60 m fixed sheet
Area per MW electrical, EOL	~4,500 m ² (Rev 1.1: ~2,500 m ² at beginning of life, normal incidence)
Blanket mass	2.5 kg/m ² ; ~13.5 t
Capacity factor	~100 %, against 20–25 % for ground solar
Storage	Safe-mode pack, order 50 kWh, for attitude, comms and heaters through a fault

7. Compute Vault and Radiation

Rev 1.1 carried 15–50 t of water and structure as shielding: a crew figure applied to electronics. Silicon needs a tolerable total dose, a handled upset rate, and a refresh plan.

Parameter	Value
Vault	Unpressurised cylinder $\varnothing 4 \times 10$ m; trays, storage pool, power conversion and loop pumps inside
Wall	10–15 mm aluminium-equivalent; cuts annual dose by 3–5× against thin structure
Total dose environment	Order of several krad/yr behind thin aluminium at 650 km polar, SAA-proton dominated; commercial accelerators and memory demonstrated tolerant into the 10–30 krad range
Tray life	3–5 years, matched to the commercial hardware cycle
Upset handling	ECC memory, frequent checkpoint to local storage, scheduling that expects a few percent throughput lost to restarts on polar and SAA passes. Open question; not closed in Rev B.
Servicing	Cold-plated trays on rails, swappable by a robotic servicer without breaking the loop; EVA-compatible interfaces; array blankets and radiator wings line-replaceable
Vault structure and shielding mass	~5 t

8. Mass Budget (Rev B, bottom-up)

Group	Basis	Mass (t)
Solar array sheet	5,400 m ² at 2.5 kg/m ²	13.5
Radiator wings and loop	2,000 m ² at 6 kg/m ²	12.2
Compute trays and storage	1 MW IT at ~12 kg/kW	12.0
Vault structure and shielding	ø4 × 10 m, 10–15 mm Al-eq, rails	5.0
Power conversion, distribution, pumps, fluid, avionics, comms	Allocation	5.0
Truss, mechanisms, attitude control, electric propulsion and propellant	Allocation	7.0
Subtotal		54.7
Growth margin, 20 %		10.9
Node mass		~65

Rev 1.1 carried 40–80 t for 200–500 kW of accelerator. Rev B carries ~65 t for 1,000 kW: the water shield and batteries are gone and compute is accounted at liquid-cooled rack density. A refresh relaunches only the ~12 t of trays and storage.

9. Arrangement

The array is a 90 × 60 m sheet in the orbit plane facing the sun. A truss runs anti-sunward from its centre; the vault sits 8–18 m behind the array, and two 50 × 20 m radiator wings extend from 20 to 70 m behind it, one zenith and one nadir of the truss, in the plane containing the sun vector and nadir. The array is edge-on to the velocity vector and shades the wings; the wings are edge-on to the sun and to the array. Truss lengths and the wing split are placeholders; areas are not. The arrangement drawing is published with the web page.

10. Economics

Both an orbital node and a terrestrial hall buy the same accelerators, on the order of \$40–50 M per MW of current-generation hardware, and those dominate either bill. What differs is the plant around them. A terrestrial AI hall costs roughly \$10–15 M per MW to build plus ~\$0.6–0.9 M per MW-

year in energy: about \$15–20 M per MW of plant and power over a five-year hardware life, before the years spent in an interconnect queue. The orbital node replaces all of it with launch mass.

Launch price to SSO	Launch cost per 65 t node	Against \$15–20 M terrestrial plant + power
\$3,000 /kg	\$195 M	10× worse
\$1,500 /kg	\$98 M	5× worse
\$500 /kg	\$33 M	~2× worse
\$300 /kg	\$20 M	parity
\$200 /kg	\$13 M	better
\$100 /kg	\$7 M	2–3× better

Crossover near \$300/kg, before on-orbit assembly, qualification and operations (which push it lower) and before the value of skipping a multi-year grid queue (which pushes it higher). Refresh at \$500/kg relaunches ~12 t for ~\$6 M against \$40 M or more of hardware inside, so refresh does not decide the case. The gate is launch price, not technology.

11. Scaling

- **Fixed unit.** Deployed area, bending modes, pointing disturbance and fault-domain size grow faster than the compute they carry. The 1 MW node is the unit; the fleet grows.
 - **Plane fill.** Nodes in one dawn-dusk plane share ground stations and crosslink; ten nodes is 10 MW with one ground segment.
 - **Independent failure.** A punctured loop or lost wing throttles one node, not the fleet.
 - **One servicer, one tray standard.**
 - **Lifecycle split.** Plant 10–15 years; trays 3–5. The plant is the asset; the trays are inventory.
-

12. Beyond Earth Orbit

Around the Moon or Mars there is no advantage until customers are there, and the Earth link is worse. The case changes only on the lunar polar surface.

Site	Thermal environment	Verdict
Lunar equatorial surface, day	~390 K ground at noon for 14 days; regolith is an insulator, conduction into the ground is negligible	Not a cold sink. Radiators see hot ground and sun, then 14 days of darkness with no power.
Lunar polar rim	Lit 80–90 % of the year at grazing angles; permanently shadowed floors at 40–100 K alongside	A vertical radiator facing the shadowed floor and cold sky rejects ~490 W/m ² one-sided at 315 K, no Earth IR, no solar load: comparable to the orbital wing per m ² .

The gain at the pole is not the sink; it is what surrounds it. Once regolith structure and local water are available the plant stops being launched, radiators can be built as walls, deployment mechanisms disappear, and the program's water provides thermal storage across the short polar nights. The LEO node is the near-term product and the testbed for the plant. The lunar polar node waits on the same gate as the rest of the program: a viable moonbase and industrial-scale lunar water.

13. Changes From Rev 1.1

Item	Rev 1.1	Rev B
Orbit	500–600 km generic; 35 % eclipse	650 km dawn-dusk SSO; no eclipse
Eclipse storage	2.6 t/MW or throttling	Safe-mode pack only
Accelerator load	200–500 kW in a 1–2 MW bus	1.0 MW IT, 1.2 MW electrical
Radiator area	1,800–3,300 m ² /MW at 290–330 K	~1,700 m ² /MW at 315 K mean; 2,000 m ² per node
Solar area	~2,500 m ² /MW BOL	~4,500 m ² /MW EOL; 5,400 m ² per node
Shielding	15–50 t water and structure	~5 t aluminium vault; 3–5 year tray refresh
Node mass	40–80 t	~65 t bottom-up, 20 % growth
Use case	Power density, thermal isolation, space adjacency	Batch training and inference; space-native data second
Data path	Not stated	Launched storage up, optical down, budget stated
Economics	Not stated	Parity near \$300/kg to SSO